

Lab Report: Hotel Network Design and Implementation

Objective

To design and implement a modern hotel network with the following requirements:

1. Three routers connecting each floor of the hotel, located in the server room.
2. Serial DCE cable connections between the routers with the network:
 - 10.10.10.0/30
 - 10.10.10.4/30
 - 10.10.10.8/30
3. Individual switches for each floor, with departments configured in different VLANs.
4. WiFi networks for laptops and phones on each floor.
5. Each department equipped with printers.
6. All devices to communicate via VLANs and IPs assigned dynamically by DHCP.
7. Use OSPF as the routing protocol to advertise routes.

Network Design Overview

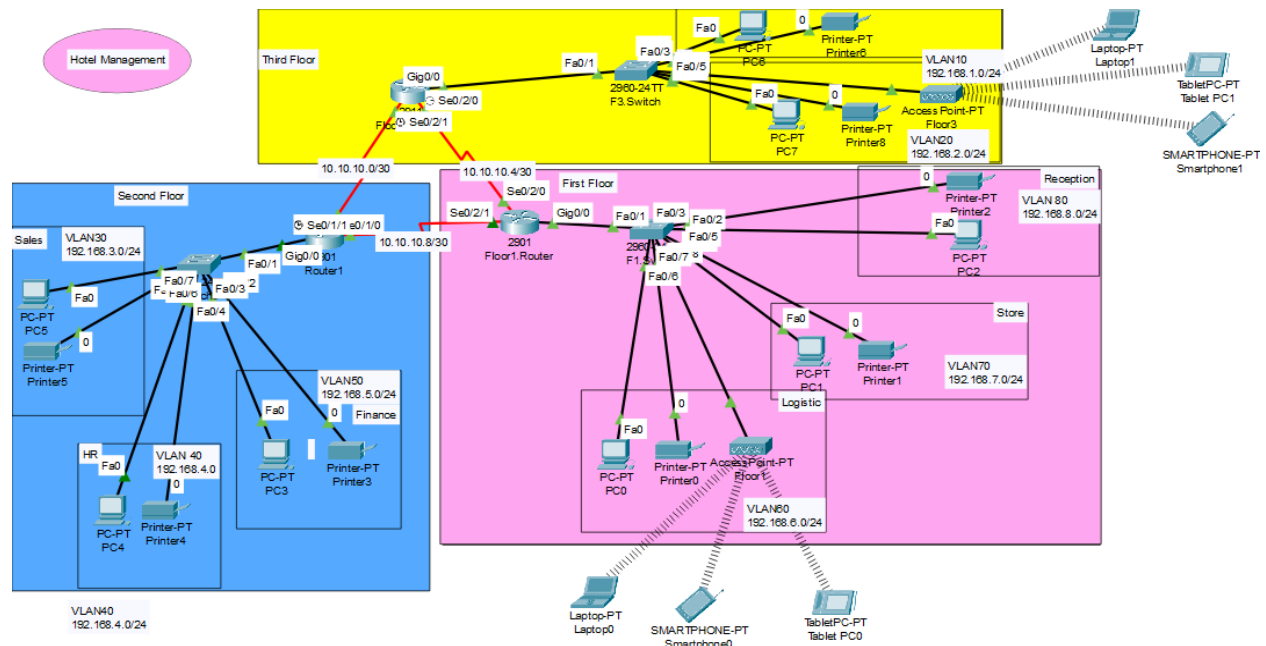
Floor and VLAN Details

1. **1st Floor:**
 - Reception: VLAN 80, Network: 192.168.8.0/24
 - Store: VLAN 70, Network: 192.168.7.0/24
 - Logistics: VLAN 60, Network: 192.168.6.0/24
2. **2nd Floor:**
 - Finance: VLAN 50, Network: 192.168.5.0/24
 - HR: VLAN 40, Network: 192.168.4.0/24
 - Sales: VLAN 30, Network: 192.168.3.0/24
3. **3rd Floor:**
 - Admin: VLAN 20, Network: 192.168.2.0/24
 - IT: VLAN 10, Network: 192.168.1.0/24

Router Connections

- Router 1 (1st Floor) ↔ Router 2 (2nd Floor) via Serial Link: 10.10.10.0/30
- Router 2 (2nd Floor) ↔ Router 3 (3rd Floor) via Serial Link: 10.10.10.4/30
- Router 1 (1st Floor) ↔ Router 3 (3rd Floor) via Serial Link: 10.10.10.8/30

Network Topology:



**Fig: Hotel Management System
Network Design**

Stick VLAN Implementation Switch

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
Switch(config-if-range)#int range fa0/4-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
Switch(config-if-range)#
Switch(config-if-range)#int range fa0/1
Switch(config-if-range)#switchport mode trunk

Switch(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if-range)#do wr
Building configuration...
[OK]
Switch(config-if-range)#
```

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Router Configuration

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#int se 0/1/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router(config-if)#int se0/1/1
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

Router(config-if)#int gig0/0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#int se0/1/1
Router(config-if)#clock rate 64000
Router(config-if)#
Router(config-if)#
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#
```

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DHCP Implementation

```

Router(config)#
Router(config)#int gig0/0.40
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#ex
Router(config)#
Router(config)#int gig0/0.50
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.50, changed state to up

Router(config-subif)#encapsulation dot1Q 50
Router(config-subif)#ip address 192.168.5.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
Router(config)#service dhcp
Router(config)#ip dhcp pool Sales
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#network 192.168.3.0 255.255.255.0
Router(dhcp-config)#
Router(dhcp-config)#default-router 192.168.5.1
Router(dhcp-config)#default-router 192.168.3.1
Router(dhcp-config)#dns-server 192.168.3.1
Router(dhcp-config)#
Router(dhcp-config)#ex
Router(dhcp-config)#
Router(config)#

```

OSPF Implementation

```

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed st
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/1, changed st
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed st

Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#router ospf 10
Router(config-router)#network 10.10.10.4 255.255.255.253 area 0
OSPF: Invalid address/mask combination (discontiguous mask)
Router(config-router)#network 10.10.10.4 255.255.255.252 area 0
Router(config-router)#network 10.10.10.8 255.255.255.252 area 0
Router(config-router)#network 192.168.8.0 255.255.255.0 area 0
Router(config-router)#network 192.168.7.0 255.255.255.0 area 0
Router(config-router)#network 192.168.6.0 255.255.255.0 area 0
Router(config-router)#
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#

```

Last Status

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC4	PC0	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC3	PC7	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC6	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC5	PC0	ICMP		0.000	N	3	(edit)	(delete)

Testing and Verification

- Use the following commands to verify configurations:
 - ping to check connectivity between devices.
 - show vlan brief to verify VLANs on switches.
 - show ip route to confirm OSPF routing.

Results

- All departments are assigned to their respective VLANs.
- Devices dynamically receive IP addresses from DHCP servers.
- Communication between floors and departments works as intended.
- OSPF successfully advertises all routes.

Conclusion

The network design meets all requirements of the modern hotel network, including VLAN segmentation, DHCP dynamic IP allocation, and inter-floor communication via OSPF. This implementation ensures efficient and secure networking for all departments.